SUMMARY · MK 6 BUSINESS ANALYTICS · WEEK 2

How to Deal with Data

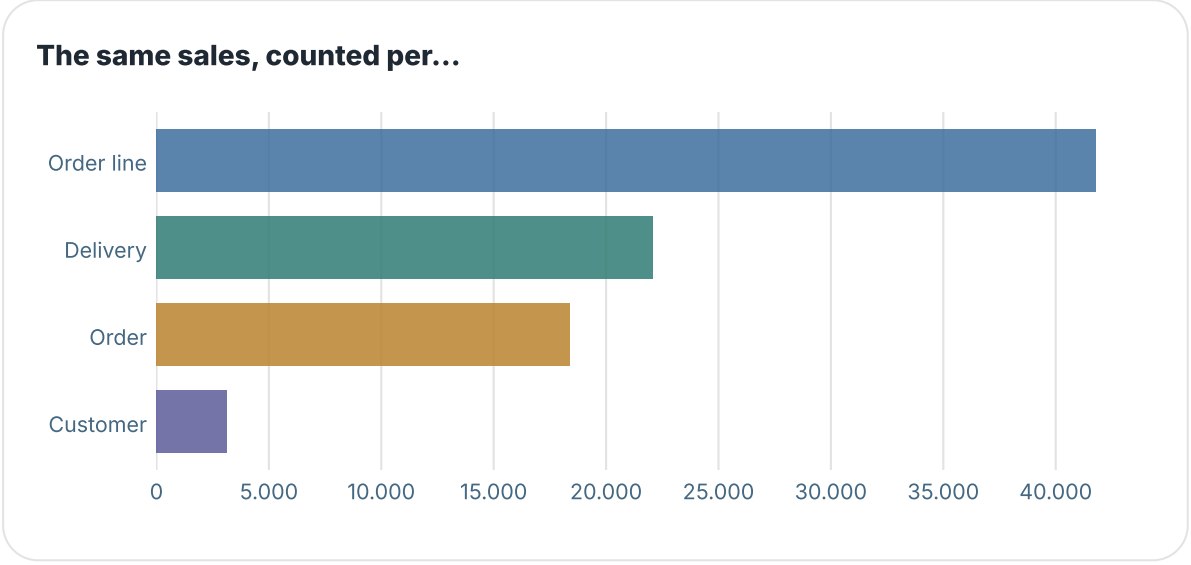
Understand data before analysing: what one row is (grain), keys, structure, metadata and data dictionaries, measurement scales, column shape (mean vs median, spread, zeros), eight data defects, and where to fix them (capture, definition, pipeline).

Material

1 One row, one thing

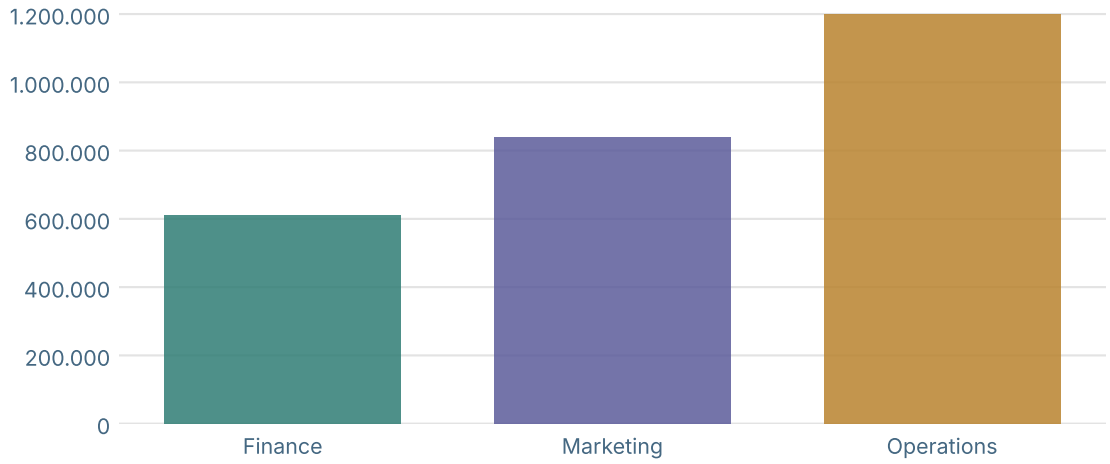
Three teams, three "active customer" counts — no one miscounted; definitions differ. Ask what one row is: order lines (41,800), orders (18,400), deliveries (22,100), customers (3,150). A key makes rows unique. Data can be structured, semi-structured or unstructured.

2 One set of sales, four row counts Deep dive



Dividing the same total by different grains answers different questions.

'Active customers' by department



No miscount — three unwritten definitions.

3 Metadata and data dictionaries

Metadata turns a number into a fact: what, where, period, source, extraction time, owner. Every column needs name, meaning, unit, source, refresh and owner — in a data dictionary the report points to.

4 Measurement scales and allowed operations

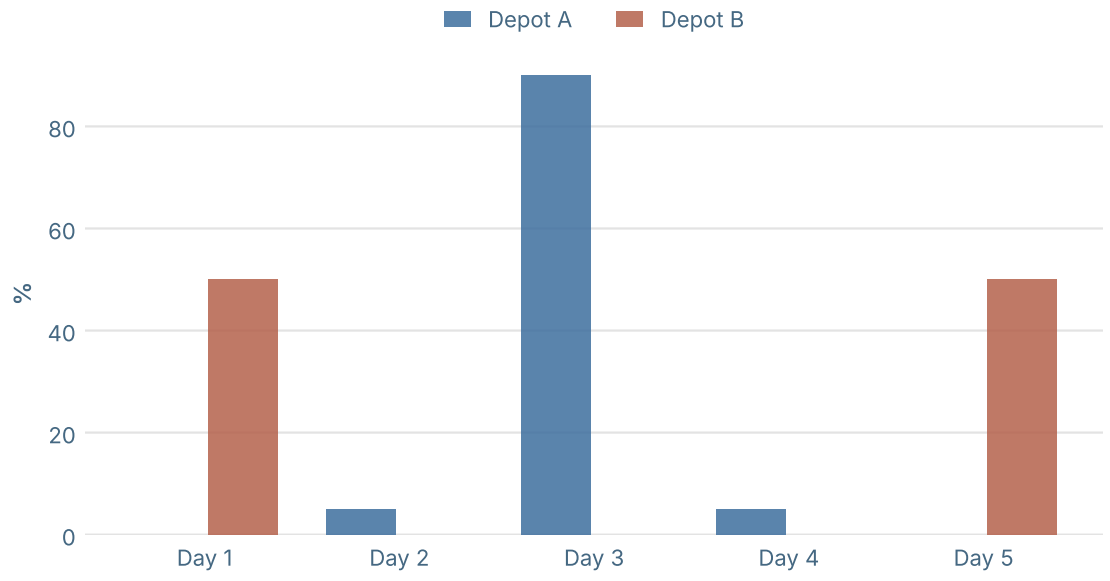
Nominal, ordinal, interval, ratio — the type decides allowed operations. Don't average branch codes or Likert scores (report share "good or better"). 80% → 84% is 4 points or 5 percent. Dates have formats, zones and pairs. Calculated columns are opinions about other columns.

5 The shape of a column

Mean 340 vs median 120 bags; same 3-day mean with different spreads; always ask for median, spread, count and a picture; count zeros (IDR 14m average hides 30 zero branches).

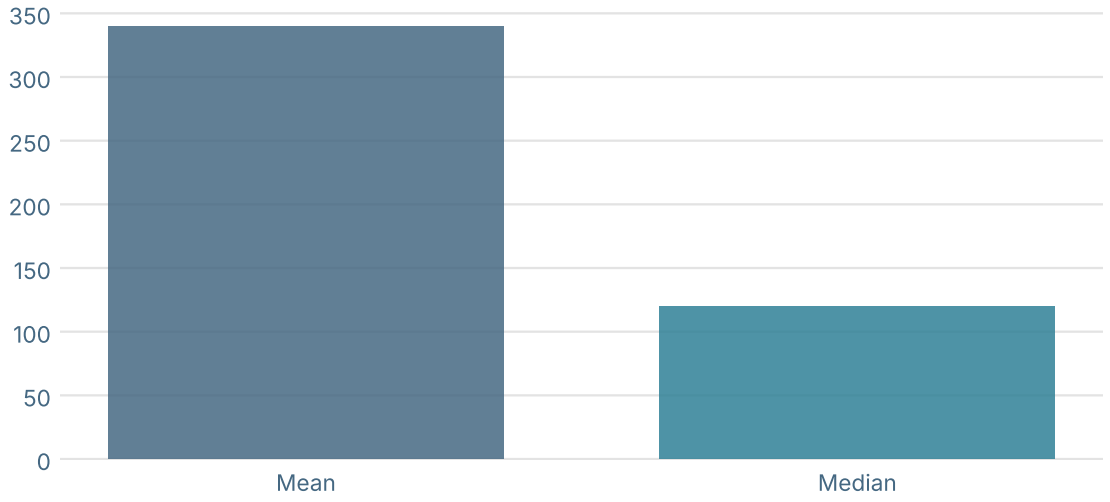
6 Column shape: averages can mislead Deep dive

Delivery-time distribution: same mean (3 days)



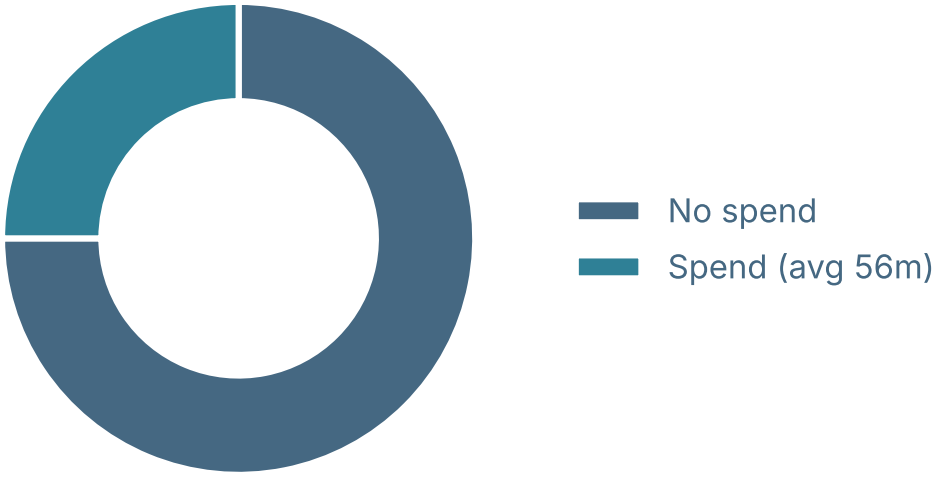
Same mean, nobody at depot B waits 3 days.

Order size: mean vs median (bags)



9 of 10 orders are below the mean.

40 branches and marketing spend



7 Eight defects and who fixes them

Eight defects: missing, duplicate, inconsistent, wrong unit, stale, wrong grain, outlier, undefined — each with a first move. Fix at capture (cheapest, process owner), definition (management) or pipeline (data team, never ends). Ask clearly for data and ask back about age, dropped rows, source and owner.

8 Three places to fix: cost vs coverage Deep dive

↕ Top row: fixes future rows · bottom: only existing rows

Capture Fix the form.	**Definition** Cheap to write, costly to agree.
—	**Pipeline** Never finished.

↔ Left column: cheap per fix · right: expensive

Key takeaways

1. What is one row?
2. Keys.
3. Metadata essentials.
4. Scales limit operations.
5. Mean needs spread.
6. Fix at capture.

Case studies

Case & worked answers

Linking: BetonPrima and 'IDR 14m per branch'

BetonPrima's churn reasons are a capture defect — fix the form. For a "3.1-day average delivery" report, ask about grain, median/P90, blanks, the definition of "delivered" and extract age.

Worked examples

Problem 1 · Easy · **Poin persentase vs persen**

Conversion rate naik dari 2,0% menjadi 2,5%. Tulis kenaikannya dengan dua cara.

1. Selisih langsung: 2,5 − 2,0 = **0,5 poin persentase**.
2. Kenaikan relatif: 0,5 ÷ 2,0 = **25%**.

Answer: Naik 0,5 poin persentase, atau naik 25%. Keduanya benar — sebutkan yang dimaksud.

Problem 2 Easy **Rata-rata vs median, dan kolom bernilai nol**

(a) Tujuh order (zak): 5, 6, 7, 8, 9, 10, 120. Hitung rata-rata dan median. (b) 50 toko, 35 tidak belanja iklan sama sekali, total belanja Rp600 juta. Hitung rata-rata semua toko dan rata-rata toko yang belanja.

1. (a) Total = 165 → rata-rata 165 ÷ 7 = 23,6. Median (nilai tengah ke-4) = 8.
2. Satu order 120 menarik rata-rata hingga tiga kali median; 6 dari 7 order di bawah rata-rata.
3. (b) Rata-rata semua = 600 ÷ 50 = 12 juta.
4. Yang belanja hanya 15 toko → 600 ÷ 15 = 40 juta.

Answer: (a) Rata-rata 23,6 vs median 8 → laporkan median. (b) Rp12 juta (semua) vs Rp40 juta (yang belanja) — hitung nol dulu sebelum membagi.

Practice questions

Try answering first. The answer key is below each question.

1. First question before reading a table...

A. Total?
B. What is one row? (grain)
C. Author?
D. Colour?

Answer: B. Grain defines the average.

2. Satisfaction 1–5 is...

A. Nominal
B. Ordinal
C. Interval
D. Ratio

Answer: B. Ordered, unequal gaps.

3. On-time rises 80% → 84%...

A. +4 points or +5%
B. +4% only
C. +84%
D. +20%

Answer: A. 4/80 = 5%.

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4. Three spellings of one company is...

A. Missing
B. Duplicate
C. Stale
D. Outlier

Answer: B. Use a stable key (tax ID).

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5. The cheapest fix covering all future rows...

A. Pipeline
B. Capture
C. Dashboard
D. Model

Answer: B. Stop bad values entering.

Sources

- Materi Lengkap P3MD — How to Deal with Data (MK6)